

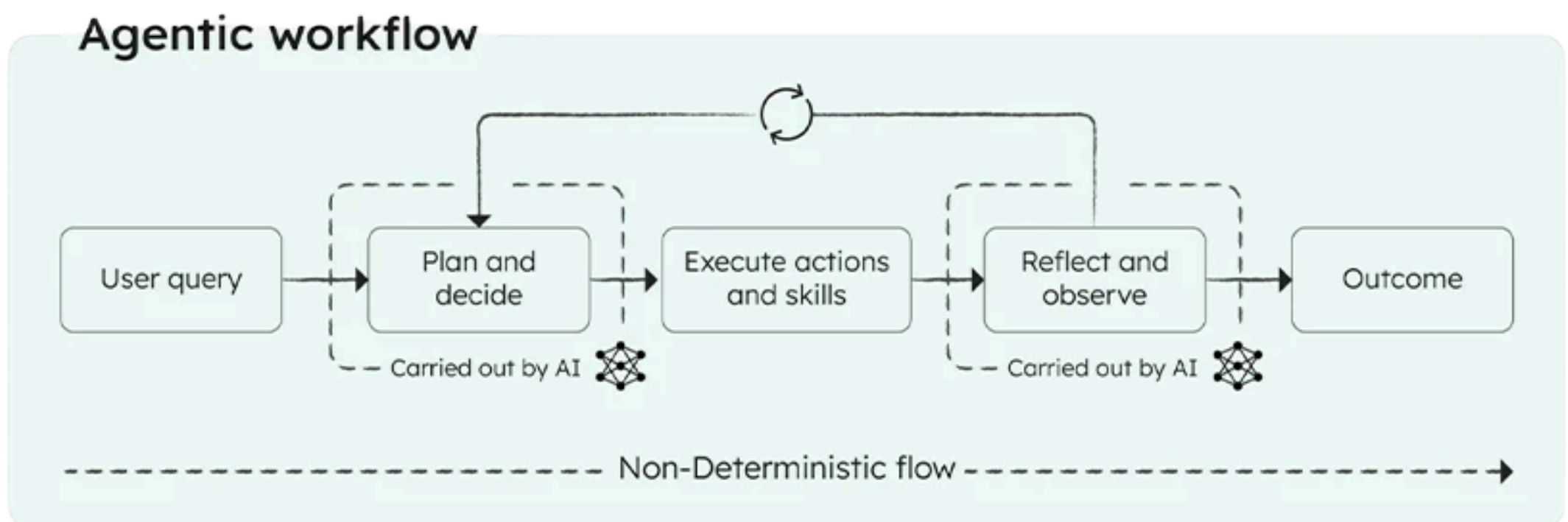
Build Reliable Agentic AI Systems

Agentic AI systems don't just respond they **plan, act, use tools, and iterate.**

That power introduces new risks: hallucinations, unsafe outputs, silent failures, and unpredictable behavior.

Reliability comes from three pillars:

- Strong foundations (architecture + memory)
- Continuous evaluation
- Tight safety + observability loops



Let's break it down

Core Architecture of Agentic Systems

A typical agentic system has these layers:

1. Planner / Reasoning Engine

- Breaks tasks into steps
- Decides which tools to use
- Uses structured planning and model reasoning to choose next actions.

2. Tool Layer

- APIs, databases, search, code execution
- Enables real-world actions

3. Execution Loop (Agent Loop)

- Think → Act → Observe → Repeat
- Stops when goal is reached or limit exceeded

4. State Manager

- Maintains context across steps
- Tracks intermediate outputs

Memory Design (Critical for Reliability)

Memory is what separates a toy agent from a production system.

Types of Memory

1. Short-Term Memory (Session Context)

- Current conversation or task state
- Stored in prompt or structured state
- Risk: context overflow → truncation issues

2. Long-Term Memory

- Persistent knowledge across sessions
- Stored in a vector database or knowledge store for retrieval, recall, and personalization
- Used for personalization or recall

3. Episodic Memory

- Past agent actions and outcomes
- Helps avoid repeating failures

4. Tool Memory

- Logs of tool usage and results
- Useful for debugging and optimization

Evaluation with DeepEval

Traditional single-turn accuracy metrics are not enough for agentic systems, so you also need behavioral evaluation.

What to Evaluate

1. Task Success

- Did the agent achieve the goal?

2. Reasoning Quality

- Logical steps
- Correct tool usage

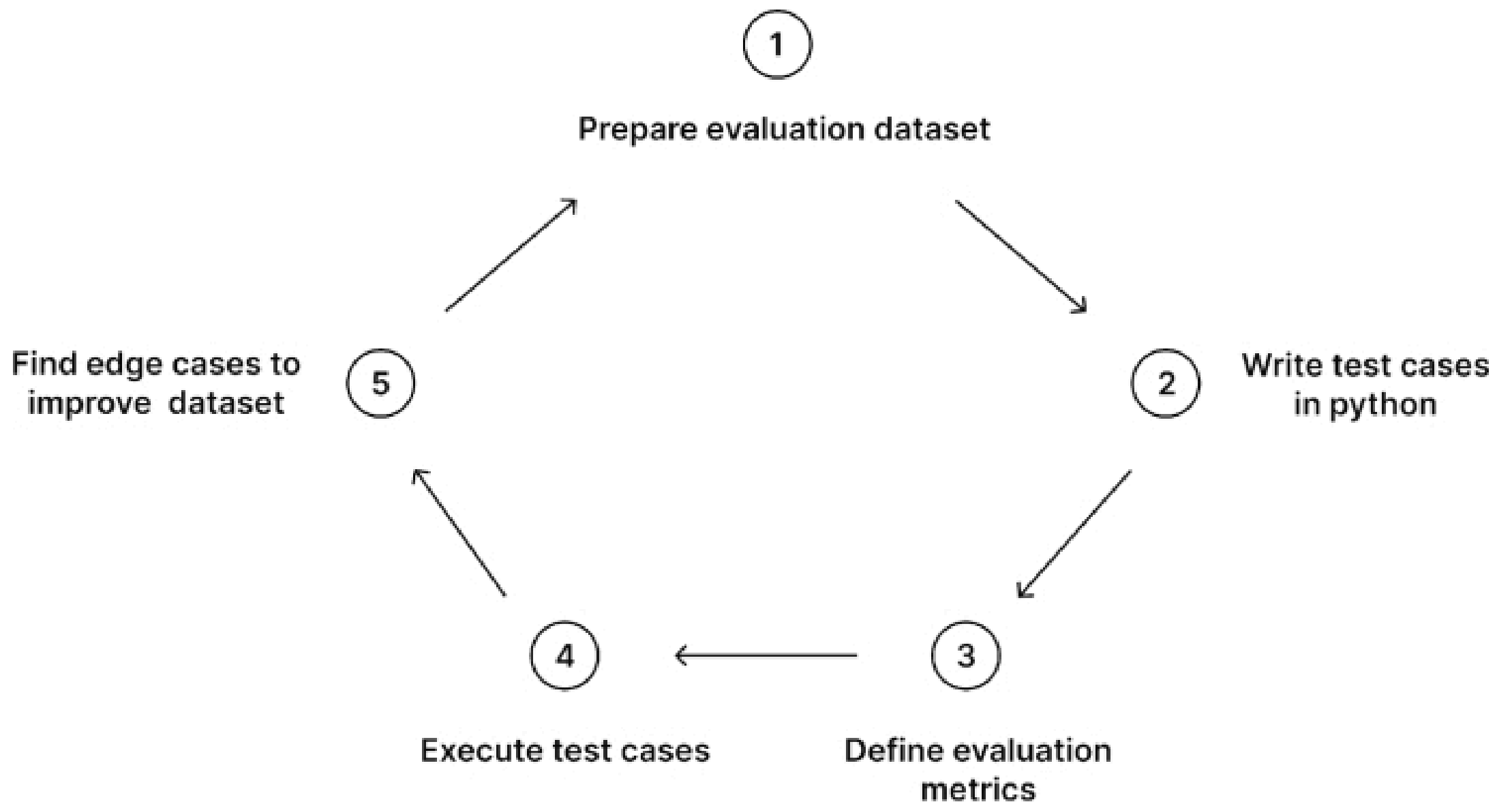
3. Faithfulness

- Is output grounded in context?

4. Safety

- No harmful or policy-violating responses

DeepEval Capabilities



With DeepEval you can:

- Run LLM-as-a-judge evaluations
- Evaluate agent behavior with:
 - Task completion
 - Tool correctness
 - Faithfulness
 - Safety
 - Multi-step workflow evaluation.

Ready to build **Reliable Agentic AI Systems**?

Join our hands-on Workshop

Build Reliable Agentic AI systems: Evaluation, Observability, and Safety



Aug 08, 2026



09:30AM – 05:30PM



Bengaluru

Workshop Modules

Module 1 : Foundations & Agent Architecture

Module 2 : Memory, MCP & Conversational Agents

Module 3 : Agent Evaluation with DeepEval

Module 4 : NeMo Guardrails & Llama Guard

Module 5 : Observability & LangSmith

Module 6 : Deploy on Hugging Face Spaces + Streamlit

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